

RIYASDEEN SAMSUDEEN

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Master's student in Human-Computer Interaction and Design with hands-on experience in electronics, embedded and IoT systems, robotics and drones particularly in user-centric product design and real-world prototyping and development, with exposure to startup environments. Passionate about leveraging technology and design thinking to solve real-world challenges and eager to contribute to cutting-edge innovations.

Portfolio Link : <https://bit.ly/riyasprofile>

RESEARCH EXPERIENCE

Aalto - Product Development Project - Finnident Oy, Finland

[Sept 2025 - May 2026]

Project Manager II and Engineer

- Led the technical redesign and system integration of a dental instrument bridge with two interactive displays, including electronics, industrial component selection and mechanical integration.
- Designed a custom PCB shield for IMX, developed prototypes, and oversaw manufacturing and integration. The project was offered a €10,000 development budget.

PROFESSIONAL EXPERIENCE

Spezar Tech Private Limited, Thanjavur, India

[Feb 2021 - Present]

Founder

- Led end-to-end development of various projects on robotics, embedded, and IoT, covering system architecture, electronics/PCB design, firmware, sensor integration, rapid prototyping and testing.
- Built and coordinated a multidisciplinary technical team and led product development from concept to prototype, with the works receiving recognition from government and international technology and entrepreneurship awards.

INRO Labs, Chennai, India

[June 2018 - Nov 2020]

Junior Research & Development Intern and STEM Trainer

- Worked on design and development of surveillance robot for COVID-19 zones.
- Trained 500+ school students on electronics, mechanical systems, product design and embedded tech.

EDUCATION

University of Twente, Netherlands

[Sept 2026 - July 2027]

***Master of Science (Technology) in ICT Innovation (EIT Digital Master's School - Double Degree) ,
Major in Human-Computer Interaction and Design with Intelligent Systems***

-Interactive technology, image processing, machine learning fundamentals, agentic & multi-agent systems, itech project - working on an interactive smart tablecloth for picky eating

Aalto University, Finland

[Sept 2025 - July 2027]

***Master of Science (Technology) in ICT Innovation (EIT Digital Master's School - Double Degree) ,
Major in Human-Computer Interaction and Design***

-Human-centred product development, user-centred design, emergent user interface with hardware, interaction design, usability & user research, digital & physical prototyping, usability engineering, cognitive psychology, innovation and entrepreneurship minors

University of the Aegean, Syros, Greece

[June 2026 - July 2026]

Summer School, Maritime Informatics & Robotics

-Maritime robotics, autonomous boats, mission planner, ardupilot, sensor integration

SASTRA Deemed to be University, Thanjavur, India

[July 2019 - May 2023]

Bachelor of Technology, Mechatronics

-Mechatronics, robotic systems, digital electronics, embedded systems, iiot, advanced control systems, electronics, sensors & instrumentation, solid & fluid mechanics, signal conditioning, mechanical design

SKILLS

Technical Skills

Product development, Industrial design, Open-Source hardware, Industrial Hardware & Sensors, PCB design, Firmware development, Sensor Integration, Image Processing, Technical Illustration, 3D modeling, 3D printing/Rapid Prototyping

Hardware Platforms

ESP, ATmega, Raspberry pi, Pixhawk, Flight Controllers, Jetson, NXP i.MX, Wireless & Wired Communication Protocols

Software Tools

Autodesk Fusion 360, Eagle, Arduino IDE, Mission Planner, Betaflight, VS Code, Bambu Studio, Cura, Canva, Figma

Programming

Embedded C/C++, Python, HTML/CSS

PROJECTS

AuraSense - Screen Free Haptic Stress Relief Device [Feb 2026 - May 2026]

- *Developed a screen-free stress-relief device combining tactile, haptic, and visual interactions as an alternative to screen-based stress and meditation tools. Contributed to electronics design, firmware, interaction logic, prototyping, and iterative user testing.*

Fully Autonomous Customer Assistive Robot for Visitor Guidance [April 2023 - Aug 2024]

- *Designed and developed a self-navigating robot platform to enhance customer service experiences and assist in retail and industrial spaces. Its a customizable mobile platform focusing seamless human-robot interaction and accessibility features. An advanced version of my old project.*

Product development for Healthcare Gait Analysis [June 2023 - Aug 2024]

- *Worked on electronics and human-centered product design of IoT based smart footwear for gait analysis, focused on diagnosing Parkinson's.*

IoT-based Organ/Medical Transportation Box [May 2023 - Aug 2023]

- *Design and Development of the hardware for secure IoT-enabled transportation box to ensure the safe and efficient delivery of organs and medical supplies through drones with real time monitoring the condition of the supplies.*

IoT System for Remote Farm Operation Integrated with VFD [Apr 2023 - Jul 2023]

- *Engineered an IoT system to remotely manage and operate farm equipment, integrated with Variable Frequency Drive (VFD) technology for efficiency. Designed a remote operating web-app dashboard with open source tools for user-friendly controls.*

Portable and Wireless Control System Module for Robotics and IoT [Jan 2023 - May 2023]

- *Designed and developed a compact and wireless control system module, enabling seamless operation of robotic and IoT devices. The system features multi-port sensor integration, ensuring flexibility, scalability, intuitive interaction, portability, and modularity.*

Portable Bio 3D Printer for Tissue Engineering Research [Feb 2022 - Jun 2022]

- *Designed and prototyped a handheld bio 3D printer as a research work, capable of fabricating tissues for enhancing research and medical capabilities. The design enhanced portability, accessibility and control the extrusion of bio gels in biomedical research.*

ACHIEVEMENTS & LEADERSHIP

- **Mentor & Innovation Ecosystem Contributor** - Mentor at Imperial Enterprise Lab's (Imperial College London) Global Challenge Lab, Evaluator at Smart India Hackathon, Volunteer at Slush Finland.
- **One of 10 Fellows from Finland** at BabyVC Nordics and Bloom Labs (TampereES) Finland.
- **Tamil Nadu Student Innovator Award** - Awarded by the Entrepreneurship Development and Innovation Institute, Government of Tamil Nadu, in 2020-21 and 2021-22.
- **International Space Challenge, Singapore** - Most Creative Award, recognized by Singapore government leadership.
- **ActInSpace** - India Top 5 in international space-tech hackathon supported by CNES, ESA and Airbus.
- **World Robotics Championship - Technoxian** - 2nd Runner-up in the Fastest Line Follower category.